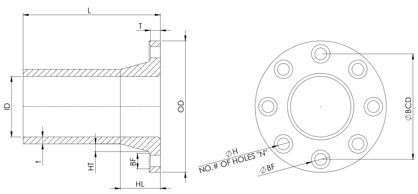
FRP Stub Flanges

½ – 36 in. SIZE RANGE

50 · 75 · 150 PSI CLASSES

12 in. STANDARD STUB LENGTH

Contact-moulded stub flanges with a full-face flat sealing surface, drilled to Class 150 through NPS 24 and ASME B16.1 Class 125 at NPS 30 and 36. Back-face diameters are spot-faced for SAE Narrow washers.



ID	nominal inside diameter	**OD**	flange outside diameter
T	flange thickness	**L**	overall stub length
t	stub wall thickness	**HT**	hub thickness at root
HL	hub length	**N**	number of bolt holes
BCD	bolt circle diameter	**H**	bolt hole diameter
BF	back-face diameter	**P**	pressure rating

ID nominal dia.	P pressure	OD flange dia.	T flange thk.	L stub length	t wall thk.	HT hub thk.	HL hub length	N holes	BCD bolt circle	H hole dia.	BF back-face
in	psi	in	in	in	in	in	in	#	in	in	in
½	150	3-5/8	7/16	12	1/4	1/4	2	4	2-3/8	5/8	1-1/4
¾	150	4	5/8	12	1/4	5/16	2-1/2	4	2-3/4	5/8	1-1/4
1	150	4-3/8	5/8	12	1/4	5/16	2-1/2	4	3-1/8	5/8	1-1/4
1-1/2	150	5-1/8	11/16	12	1/4	3/8	2-3/4	4	3-7/8	5/8	1-1/4
2	150	6-1/8	11/16	12	1/4	3/8	2-3/4	4	4-3/4	3/4	1-1/2
2-1/2	150	7-1/8	13/16	12	1/4	7/16	3-1/4	4	5-1/2	3/4	1-1/2
3	150	7-5/8	13/16	12	1/4	7/16	3-1/4	4	6	3/4	1-1/2
4	150	9-1/8	15/16	12	1/4	1/2	3-3/4	8	7-1/2	3/4	1-1/2
6	50	11-1/8	11/16	12	1/4	5/16	3-13/16	8	9-1/2	7/8	1-3/4
6	150	11-1/8	1-1/16	12	3/8	9/16	4-1/4	8	9-1/2	7/8	1-3/4
8	50	13-5/8	13/16	12	1/4	3/8	4-3/8	8	11-3/4	7/8	1-3/4
8	150	13-5/8	1-1/4	12	7/16	5/8	5	8	11-3/4	7/8	1-3/4
10	50	16-1/8	15/16	12	1/4	1/2	4-3/8	12	14-1/4	1	2
10	150	16-1/8	1-7/16	12	1/2	3/4	5-3/4	12	14-1/4	1	2
12	50	19-1/8	1-1/16	12	1/4	5/8	4-15/16	12	17	1	2
12	150	19-1/8	1-3/4	12	5/8	7/8	7	12	17	1	2
14	50	21-1/8	1-1/16	12	5/16	11/16	5	12	18-3/4	1-1/8	2-1/4
14	150	21-1/8	1-7/8	12	3/4	15/16	7-1/2	12	18-3/4	1-1/8	2-1/4
16	50	23-5/8	1-3/16	12	5/16	3/4	5-1/16	16	21-1/4	1-1/8	2-1/4
16	150	23-5/8	2	12	13/16	1	8	16	21-1/4	1-1/8	2-1/4
18	50	25-1/8	1-1/4	12	3/8	3/4	5-1/2	16	22-3/4	1-1/4	2-1/2
18	150	25-1/8	2-1/8	12	15/16	1-1/16	8-1/2	16	22-3/4	1-1/4	2-1/2
20	75	27-5/8	1-5/8	12	1/2	13/16	6-1/2	20	25	1-1/4	2-1/2
24	75	32-1/8	1-7/8	12	5/8	15/16	7-1/2	20	29-1/2	1-3/8	2-3/4
30	75	38-7/8	2-1/4	12	1	1-1/8	9	28	36	1-3/8	2-3/4
36	75	46-1/8	2-1/2	12	1-1/4	1-1/4	10	32	42-3/4	1-5/8	3-1/4

NOTES

- Drilling compatible with Class 150: ASME B16.5 for NPS ½–24, ASME B16.1 Class 125 for NPS 30 and 36.
- Bolt patterns compatible with ASME RTP-1 vessel nozzle drilling; fabrication to CGSB 41-GP-22.
- Contact moulded (hand lay-up) to the layer structure on page 03 and to ASTM C582. Supplied to ASTM D5421.
- Back-face (BF) dimensions suit ASME B18.22.1 Type A Narrow washers.

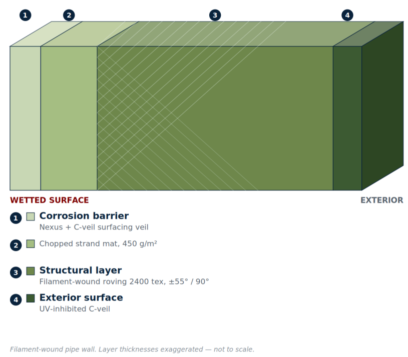
TOLERANCES

- OD: +½ in. / −0 in. · BCD: ±0.06 in. · Hole spacing: ±0.03 in.
- Eccentricity to nozzle centre: ±0.03 in. (NPS ≤2), ±0.06 in. (NPS ≥3).
- Flange face flatness: ±1/32 in. All thicknesses include the corrosion barrier.

Resins, Laminate & Standards

Every component is built as a layered laminate. The layer names below are those used by ASTM C582 and ASME RTP-1 and are the common vocabulary of the industry. What differs between products is not the structure but the ply build-up and the way the structural layer is laid down. Wall thicknesses quoted throughout this catalogue are total thicknesses and include the corrosion barrier.

LAMINATE CONSTRUCTION



Filament-wound pipe wall. Layer thicknesses not to scale.

Inner surface
A resin-rich layer reinforced with a surfacing veil — glass, synthetic or carbon. Nominally 0.010–0.020 in. This is the surface the process fluid touches.

Interior layer
Chopped strand mat, two plies minimum. With the inner surface it forms the **corrosion-resistant barrier** — the part of the wall that does the chemical work, and the part excluded from the strength calculation.

Structural layer
Carries the pressure and the mechanical load. Filament wound on pipe; contact moulded — hand lay-up — on every fitting. The first ply against the barrier is always mat.

Outer surface
A resin coat, paraffinated for a full surface cure. UV-inhibited exterior standard on pipe, available on fittings.

Ply-by-ply build-up differs by product and method. The filament-wound pipe schedule is tabulated on page 07; fitting schedules are held by the manufacturer — see the scope note below.

RESIN SYSTEMS

Derakane 411-350
Bisphenol-A epoxy vinyl ester

HDT	105 °C / 220 °F	TENSILE	86 MPa / 12,000 psi
MODULUS	3,200 MPa	ELONGATION	5–6 %

The standard offering. Broad resistance to acids, alkalis, bleaches and solvents.

Derakane 470-300
Epoxy novolac vinyl ester

HDT	150 °C / 300 °F	TENSILE	85 MPa / 12,500 psi
MODULUS	3,600 MPa	ELONGATION	3–4 %

Specified where elevated temperature, solvent or oxidiser service (chlorine, flue gas) demands higher thermal and chemical resistance.

HDT is a material property, not a service rating. Maximum service temperature is governed by the chemical environment — confirm against the resin manufacturer's chemical resistance guide before specifying.

STANDARDS OF MANUFACTURE AND REFERENCE

CGSB 41-GP-22	Process equipment — reinforced polyester, glass fibre. Governing Canadian fabrication standard; covers tanks, pipe and ductwork, and establishes the 10:1 safety factor.	**ASTM D2996**	Filament-Wound "Fiberglass" Pipe. Governs the pipe on page 07.
ASTM D5421	Contact-Molded "Fiberglass" Flanges. Governs the stub flanges and manway flanges.	**ASTM C582**	Contact-Molded Reinforced Thermosetting Plastic Laminates. Governs the laminate in all contact-moulded fittings.
ASME RTP-1	Referenced for bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns only — see the scope note below.	**ASME B16.5**	Class 150 bolt-hole drilling, NPS ½ through 24.
ASME B16.1 Cl. 125	Bolt-hole drilling, NPS 30 and 36.	**ASME B16.9**	Geometric basis for long-radius elbows (R = 1.5 × ID).
ASME B18.22.1	Type A Narrow plain washers — governs back-face (BF) diameters.	**ASME B1.20.1**	Taper pipe threads — FNPT couplings.

DESIGN BASIS

- Minimum **10:1 safety factor** on the structural laminate, per CGSB 41-GP-22.
- The corrosion-resistant barrier is excluded from the strength calculation — the safety factor applies to the structural wall only.
- Filament-wound structural layers are designed on a **2:1 hoop-to-axial** stress ratio.

GASKETS & BOLTING

- 1/8 in. full-face elastomeric gasket, Shore A60 ± 5, minimum design seating stress (y) 100 psi, gasket factor (m) 1.
- Flat-face mating flanges only — never bolt FRP to a raised-face metallic flange without a full-face adaptor.
- Tighten in a star pattern in equal stages. FRP flanges are damaged by over-torquing — ask us for torque values.

STANDARDS SCOPE

- **ASME RTP-1 is a vessel standard** rated to 15 psig. It does not cover piping or pipe fittings. Where we cite it, we mean bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns — nothing here is offered as an RTP-1 certified vessel component.
- We are a stockist. Item-specific laminate schedules and conformance documentation come from the manufacturer against a specific order — ask at enquiry stage.

FRP DEPOTS

CANADA’S DIRECT SOURCE FOR FRP COMPONENTS

How to order

Send your enquiry to **sales@frpdepots.com** or call **613-499-4950**. Include the following and we will quote from stock where we can.

01 Product type and nominal diameter

02 Pressure rating required

03 Resin system — D411 or D470

04 Quantity and length (pipe, couplings)

05 Any non-standard drilling, stub length or joint preparation

06 Drawings or job specification, if you have them

> Full dimensional data, tolerances and design basis are published in **FRPD-SPEC-001 Revision R3**. Request the current revision with your enquiry.

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